ASSIGNMENT 1 **(CLO1-PLO2-C4)**

Course Name: Control Systems

Course Code: EEE319

Learning Outcome: Analyse the stability, accuracy, and response speed of control systems manually **(CLO1).**

Assessment Weightage: 15% of the total course marks

Total Marks: 100

Due Date: **20 Jun 2025**

Scenario:

You have been appointed as a control systems analyst at a manufacturing plant. The management is reviewing the performance of the plant's motor control system modelled by the transfer function:

$$G(S) = \frac{10}{s^2 + 4s + 10}$$

Your task is to manually analyse the system for stability, accuracy, and response speed.

Tasks:

1. **Stability Analysis** **(Total 30 Marks)**
   a) Use the **Routh-Hurwitz Criterion** to determine the stability of the system.

   (15 marks)

   b) Sketch the **pole-zero plot** of the system and explain how the pole locations relate to system stability.

   (15 marks)

2. **Response Speed** **(Total 30 Marks)**
   a) Calculate **natural frequency** (ω_n) and **damping ratio** (ζ) from the characteristic equation.

   (10 marks)

b) Calculate and discuss the system's **rise time**, **peak time**, and **settling time** for a unit step input.

(15 marks)

c) Sketch the expected **unit step response** curve based on your manual calculations.

(5 marks)

3. Accuracy Analysis (Total 20 Marks)

a) Classify the system type and determine the **steady-state error (SSE)** for a unit step, ramp, and parabolic input using error constants.

(15 marks)

b) Comment on the system's ability to track different types of inputs

(5 marks)

4. Recommendations & Conclusion (Total 20 Marks)

a) Based on your analysis, suggest one modification (e.g., adding a zero or a compensator) to improve the accuracy or speed of the system. Support your suggestion with reasoning.

(10 marks)

b) Summarise your key findings in terms of the system's performance and reliability.

(10 marks)

APPENDIX 1

Rubric:

Assessment Criteria	Excellent (Full Marks)	Good (75%)	Satisfactory (50%)	Poor (25%) or Incomplete	Marks
Stability Analysis	Correct and complete Routh table, accurate stability interpretation; well-drawn pole-zero plot with strong justification.	Minor mistakes in calculation or interpretation.	Attempted but contains several calculation/logic errors.	Incomplete or incorrect Routh table; poor justification.	/30
Time-Domain Response (Speed)	Correct ωn, ζ, Tr, Tp, Ts with justifications and neat sketch of step response.	Mostly correct values; acceptable sketch.	Partial values correct; sketch unclear or missing.	Incorrect or missing analysis.	/30
Accuracy Analysis	System type correctly identified; correct SSE for all inputs; clear tracking analysis.	Minor errors in SSE or explanation.	Attempted with major conceptual errors.	Wrong or incomplete analysis.	/20
Recommendations & Conclusion	Insightful improvement; well-written summary of findings.	Valid improvement; decent conclusion.	Limited analysis; vague suggestion.	No valid conclusion or irrelevant suggestion.	/20

Note to students: Please include the marking rubric when submitting your coursework.